

**DISCLAIMER:** This document contains the "whiteboard notes" taken during the "Introduction to FEM for Helmholtz" at the waves Summer School in August 2025.

Therefore, they are not complete and contain some simplifications for the sake of time, etc. Moreover, the document has not been proof-read and may therefore have some minor mistakes/typos that are typical during lectures

In the coming days I will add the corresponding references and sources for the pictures that are not my own.

### LECTURE 3: NUMERICAL CHALLENGES FOR HELMHOLTZ

What does this mean?

Quasi-optimality estimate

$$\|u - u_N\|_X \leq (1 + C_0) \inf_{w \in X_N} \|u - w\|_X$$

where  $C_0 = \frac{C_A}{\alpha_A} \rightarrow$  continuity constant  
 $\alpha_A \rightarrow$  Gårding's ineq. constant

But what Gårding's ineq?

$$\begin{aligned} a(u, u) &= \|u\|_{H^1(\Omega)}^2 - K^2 \|u\|_{L^2(\Omega)}^2 \\ &= \|u\|_{H^1(\Omega)}^2 - (1 + K^2) \|u\|_{L^2(\Omega)}^2 \end{aligned}$$

so essentially,  $\alpha_A$  depends on  $K$ .

We also said that we had convergence if

$N \geq N_0$  for a given  $N_0$

but we need  $K$ -explicit information about the

Helmholtz solution to quantify how  $N_0$  and  $C_0$  depend on  $K$ .

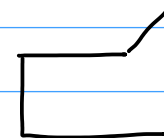
### 3.1 Trapping (why we like it and fear it)

In the singular limit  $K \rightarrow \infty$ , the wave transitions to a particle / ray billiards model

So, it is different to consider a

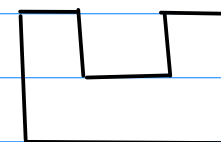
• non-trapping domain

(for the exterior problem)



• trapping domain

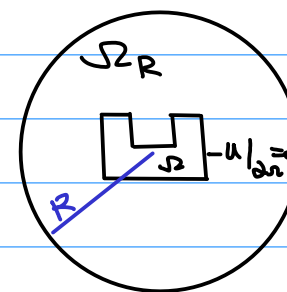
Supports a trapped ray / billiard trajectory.



① How does this affect the numerics?

The tool to figure this out is

Resolvent estimates.



$f$  supported in  $\Omega_R$

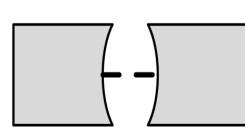
$$-\Delta u - k^2 u = f$$

$u$  satisfies Sommerfeld rad. condition

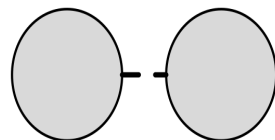
$$\|\nabla u\|_{L^2(\Omega_R)} + K \|u\|_{L^2(\Omega_R)} \lesssim c(K) \|f\|_{L^2(\Omega_R)}$$

Trapping also happens when considering scattering at multiple domains

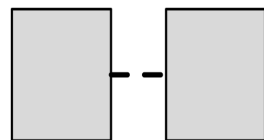
Types of trapping



elliptic



hyperbolic



parabolic

(These names have nothing to do with the PDE)

Then, we have

|                         |                       |              |  |
|-------------------------|-----------------------|--------------|--|
| non-trapping            | $c(k) = 1$            |              |  |
| elliptic                | hyperbolic            | parabolic    |  |
| $c(k) = \exp(\alpha k)$ | $c(k) = \log(2\pi k)$ | $c(k) = k^2$ |  |

### 3.2 Pollution effect

Back to our quasi-optimality estimate

$$\|u - u_N\|_X \leq (1 + C_0) \inf_{w \in X_N} \|u - w\|_X \quad (E1)$$

We want to know how  $C_0$  and  $N_0$  depend on  $K$ .

For  $h$ -FEM we need  $\dim(X_N) \sim K^d$

but  $\dim(X_N) \gg K^d$  for (E1) to hold

This extra-requirement is called "pollution effect"

$h$ -BEM,  $\Sigma$  non-trapping you have no pollution

if  $\dim(X_N) \sim K^{d-1}$

For  $hp$ -FEM,  $\Sigma$  non-trapping  $\dim(X_N) \sim K^d$   
 $hp$ -BEM

Provided that  $p \sim \log(K)$ .